

A prefix deficit law for the off-hook order law: each prefix block consumes exactly one level of sign-kill (the family $\lambda = (2, 2, 1^m)$, uniformly in m)

Clio

30 May 2026 (prove session)

Abstract

The strong per-subset operator vanishing $M(S) \equiv 0$ on V^λ for $|S| < D = \tau(\tau + 1)/2$ governs the off-hook order law $\text{ord}_{x=q^2} Z_\lambda = D$. The companion paper “*A deficit law for Strong Pillar 1*” (29 May 2026) proved the *tail* half: the tail chain produces sign-kill depth $\tau - |S_T|$, i.e. $M_T(S_T)V^\lambda \subseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-$ for $|S_T| < \tau$. Its closing Remark identified the missing piece as the *prefix counterpart*, controlling $M_R(S_R)$. We supply it. Writing $M(S) = M_R(S_R)M_T(S_T)$, the prefix acts on the tail’s sign-kill image, and we prove that *each prefix block consumes exactly one level of sign-kill*, at a price of one P_{-1} -insertion per surviving level. The mechanism is two elementary facts about the Hoefsmit seminormal module: a sign-kill subspace at depth d forces the first d letters of a block to be P_{-1} (else the block annihilates it), and the remaining block letters — carrying generators of index $\geq d + 1$ — commute past the generators $1, \dots, d - 1$ (axial gap ≥ 2) and so preserve depth $d - 1$. Iterating across the prefix blocks gives the **Prefix Deficit Law**: for $\lambda = (2, 2, 1^m)$, if $|S_T| < \tau$ then

$$M(S) \neq 0 \implies |S_R| \geq \binom{\tau - |S_T| + 1}{2}.$$

Two corollaries follow uniformly in m : (i) on the slice $S_T = \emptyset$ the full order-law lower bound, $M(S) = 0$ for every prefix-supported S with $|S| < D$; and (ii) $\text{ord}_{x=q^2} Z_\lambda \geq \tau$. We also record, and dispose of, the crossing-symmetry route: $M(x)M(q^2/x) = I$ yields word-reversal invariance and a free per-subset adjoint flip, but neither transports the long-block tail law onto the short-block prefix, so the prefix law must be proved directly — as it is here. All structural steps are pure seminormal algebra; computations (exact rational, $q = 5/7$) confirm the law with no violation for $m = 3, 4$.

1 Setup

We use the conventions of the 29 May deficit paper. $H_n(q)$ is the Iwahori–Hecke algebra, $T_i^2 = (q - 1)T_i + q$; on the Hoefsmit seminormal module V^λ set $S_i := T_i + 1$ and

$$E_i^+ := \ker(T_i - q)|_{V^\lambda} = \text{im}(P_q^{(i)}), \quad E_i^- := \ker(T_i + 1)|_{V^\lambda} = \text{im}(P_{-1}^{(i)}),$$

where $P_q^{(i)} = (T_i + 1)/(q + 1)$ and $P_{-1}^{(i)} = (q - T_i)/(q + 1)$ are the orthogonal eigenprojectors, $P_q^{(i)} + P_{-1}^{(i)} = \text{id}$, $E_i^- = \ker P_q^{(i)}$.

The staircase word and the degenerate monodromy. For $\lambda = (2, 2, 1^m)$ we have $n = m + 4$, $N = \binom{n}{2}$, $\ell(\lambda) = m + 2$,

$$\tau := \tau(\lambda) = m - 1, \quad D := \frac{\tau(\tau+1)}{2} = \binom{m}{2}.$$

Write $B_b := S_b S_{b-1} \cdots S_1$ for the descending block of length b (generators $1, \dots, b$), $b = 1, \dots, n-1$. The full staircase operator is $\Omega = B_1 B_2 \cdots B_{n-1}$. For a subset $S \subseteq \{1, \dots, N\}$ of word positions the per-subset operator is

$$M(S) := \prod_{k=1}^N A_k, \quad A_k = P_{-1}^{(i_k)} \ (k \in S), \quad A_k = P_q^{(i_k)} \ (k \notin S),$$

(i_k) the staircase letter sequence. As established (operator reformulation), $\text{ord}_{x=q^2} Z_\lambda = \min\{|S| : M(S) \neq 0\}$, so the order law is the assertion that this minimum equals D . We prove a lower bound on it.

Prefix/tail split. Following the deficit paper, the tail is the last two (longest) blocks $B_{n-2} B_{n-1}$ and the prefix is $B_1 B_2 \cdots B_{n-3}$. Split $S = S_R \sqcup S_T$ into prefix and tail positions, and factor

$$M(S) = M_R(S_R) M_T(S_T), \quad M_R(S_R) = \prod_{k \in \text{prefix}} A_k, \quad M_T(S_T) = \prod_{k \in \text{tail}} A_k.$$

Because operators act on the right factor first, $M_T(S_T)$ acts on V^λ *first* and $M_R(S_R)$ *afterwards*. The tail deficit law (Corollary 7 of the deficit paper) supplies the input to the prefix:

$$|S_T| < \tau \implies M_T(S_T) V^\lambda \subseteq \bigcap_{j=1}^{\delta} E_j^-, \quad \delta := \tau - |S_T| \geq 1. \quad (1)$$

Depth. For a subspace $W \subseteq V^\lambda$ put $\text{dep}(W) := \max\{d \geq 0 : W \subseteq \bigcap_{j=1}^d E_j^-\}$ (with $\text{dep}(W) = 0$ if $W \not\subseteq E_1^-$). Thus (??) reads $\text{dep}(M_T(S_T) V^\lambda) \geq \delta$.

2 The prefix consumes sign-kill, one level per block

The whole content is two lemmas about how a single descending block B_b acts, by left multiplication, on a sign-kill subspace. Recall $B_b = S_b S_{b-1} \cdots S_1$, so when applied to a vector the rightmost letter S_1 acts first, then S_2 , ..., then S_b ; with per-subset projector choices the block becomes $O_b = P^{(b)} P^{(b-1)} \cdots P^{(1)}$, $P^{(i)} \in \{P_q^{(i)}, P_{-1}^{(i)}\}$.

Lemma 1 (A block at depth d costs d insertions). *Let $W \subseteq V^\lambda$ with $\text{dep}(W) = d \geq 1$, and let $O_b = P^{(b)} \cdots P^{(1)}$ be a block of length $b \geq d$ acting on W . If $P^{(j)} = P_q^{(j)}$ for some $j \leq d$, then $O_b W = 0$. Consequently $O_b W \neq 0$ forces $P^{(1)} = \cdots = P^{(d)} = P_{-1}$, i.e. at least d of the block's letters carry P_{-1} ; and then $P^{(d)} \cdots P^{(1)} W = W$.*

Proof. Apply the letters in acting order $P^{(1)}, P^{(2)}, \dots$. Inductively suppose $P^{(1)}, \dots, P^{(j-1)}$ are all P_{-1} and the running image is still W (true at $j = 1$). Since $j \leq d = \text{dep}(W)$ we have $W \subseteq E_j^- = \ker P_q^{(j)} = \text{im } P_{-1}^{(j)}$. Thus if $P^{(j)} = P_q^{(j)}$ then $P_q^{(j)} W = 0$ and the whole product vanishes; while if $P^{(j)} = P_{-1}^{(j)}$ then $P_{-1}^{(j)} W = W$ (a projector restricted to its image is the identity), and the running image is unchanged. Hence a nonzero outcome forces $P^{(1)} = \cdots = P^{(d)} = P_{-1}$, in which case $P^{(d)} \cdots P^{(1)} W = W$. $\square$

Lemma 2 (A block drops depth by at most one). *Let $W \subseteq V^\lambda$ with $\text{dep}(W) = d \geq 1$ and let O_b ($b \geq d$) be a block with $O_b W \neq 0$. Then $\text{dep}(O_b W) \geq d - 1$.*

Proof. By Lemma ?? the first d letters are P_{-1} and leave W unchanged, so $O_b W = P^{(b)} \dots P^{(d+1)} W$ (the letters of index $d + 1, \dots, b$, if any; for $b = d$ there are none and $O_b W = W$, giving depth $d \geq d - 1$). Each remaining letter carries a generator of index $i \in \{d + 1, \dots, b\}$. For every $j \in \{1, \dots, d - 1\}$ the axial gap satisfies $|i - j| \geq (d + 1) - (d - 1) = 2$, so T_i commutes with T_j , hence $P^{(i)}$ (a polynomial in T_i) commutes with $P_{-1}^{(j)}$ and therefore preserves $E_j^- = \text{im } P_{-1}^{(j)}$. Thus each $P^{(i)}$, $i \geq d + 1$, maps $\bigcap_{j=1}^{d-1} E_j^-$ into itself. Since $W \subseteq \bigcap_{j=1}^d E_j^- \subseteq \bigcap_{j=1}^{d-1} E_j^-$, applying $P^{(d+1)}, \dots, P^{(b)}$ in turn keeps the image inside $\bigcap_{j=1}^{d-1} E_j^-$. Hence $O_b W \subseteq \bigcap_{j=1}^{d-1} E_j^-$, i.e. $\text{dep}(O_b W) \geq d - 1$. $\square$

Remark 3. Lemma ?? is sharp: the layer E_d^- is generally lost, because the letter S_{d+1} sits at axial gap 1 from S_d and does not preserve E_d^- . Numerically the surviving depth is exactly $d - 1$ for the minimal (P_{-1} only where forced) block.

3 The Prefix Deficit Law

We iterate the two lemmas across the prefix blocks. In acting order the prefix blocks are $B_{n-3}, B_{n-4}, \dots, B_1$ (the block B_{n-3} , nearest the tail, acts first). Their lengths are $n - 3, n - 4, \dots, 1$, and block B_{n-2-k} (the k -th prefix block, $k = 1, \dots, n - 3$) carries generators $1, \dots, n - 2 - k$.

Theorem 4 (Prefix Deficit Law). *For $\lambda = (2, 2, 1^m)$ and any subset S with $|S_T| < \tau$,*

$$M(S) \neq 0 \implies |S_R| \geq \binom{\delta + 1}{2} = \binom{\tau - |S_T| + 1}{2}.$$

Equivalently, $M(S) = 0$ whenever $|S_T| < \tau$ and $|S_R| < \binom{\tau - |S_T| + 1}{2}$.

Proof. Suppose $M(S) \neq 0$; then in particular no prefix prefix-of-the-word annihilates the running image, so the image $W_0 := M_T(S_T)V^\lambda$ is nonzero and survives every prefix block. By (??), $\text{dep}(W_0) \geq \delta$.

Let W_k denote the (nonzero) image after the first k prefix blocks have acted, W_0 as above. We claim, for $k = 1, \dots, \delta$: the k -th prefix block B_{n-2-k} forces at least $\text{dep}(W_{k-1})$ distinct P_{-1} -insertions, and $\text{dep}(W_k) \geq \text{dep}(W_{k-1}) - 1$; combined with $\text{dep}(W_0) \geq \delta$ this gives $\text{dep}(W_{k-1}) \geq \delta - k + 1$.

The only hypothesis Lemmas ??-?? require of the k -th block is that its length be at least the entering depth $\text{dep}(W_{k-1})$, so that the generators $1, \dots, \text{dep}(W_{k-1})$ all occur in it. At step $k \leq \delta$ the inductive bound gives entering depth $\text{dep}(W_{k-1}) \leq \delta - k + 1$, while block B_{n-2-k} has length $n - 2 - k$. Since $\delta \leq \tau = m - 1$ and $n - 3 = m + 1$, we have $n - 2 \geq \delta + 1$, hence

$$n - 2 - k \geq (\delta + 1) - k \geq \delta - k + 1 \geq \text{dep}(W_{k-1}),$$

so the block contains the needed generators. Lemma ?? then forces its first $\text{dep}(W_{k-1})$ letters to be P_{-1} (that many distinct insertions), and Lemma ?? gives $\text{dep}(W_k) \geq \text{dep}(W_{k-1}) - 1$.

The insertions counted in different prefix blocks are distinct (they lie in different blocks), so summing over $k = 1, \dots, \delta$,

$$|S_R| \geq \sum_{k=1}^{\delta} (\delta - k + 1) = \sum_{j=1}^{\delta} j = \binom{\delta + 1}{2} = \binom{\tau - |S_T| + 1}{2}.$$

(After $k = \delta$ blocks the depth bound reaches 0 and no further insertions are forced; the remaining prefix blocks $B_{n-3-\delta}, \dots, B_1$ may be passed with all P_q .) This is the claim; the contrapositive is the displayed vanishing statement. $\square$

The block-length bookkeeping in the proof only uses $\delta \leq \tau = m - 1$ and $n - 3 = m + 1$, so there are always $\geq \tau$ prefix blocks of sufficient length to host the demanded P_{-1} 's; the argument is uniform in m .

Corollary 5 (Order-law lower bound on the prefix slice). *For $\lambda = (2, 2, 1^m)$, $M(S) = 0$ for every S supported in the prefix ($S_T = \emptyset$) with $|S| < D$. Equivalently, on the slice $S_T = \emptyset$ the first nonzero $M(S)$ occurs at $|S| = D$.*

Proof. Take $S_T = \emptyset$, so $\delta = \tau$ and Theorem ?? gives $M(S) \neq 0 \Rightarrow |S_R| \geq \binom{\tau+1}{2} = D$. Hence $|S| = |S_R| < D$ forces $M(S) = 0$. $\square$

Corollary 6 ($\text{ord} \geq \tau$). *For $\lambda = (2, 2, 1^m)$, $M(S) = 0$ for all $|S| < \tau$; hence $\text{ord}_{x=q^2} Z_\lambda \geq \tau$.*

Proof. If $|S| < \tau$ then $|S_T| \leq |S| < \tau$, so Theorem ?? applies with $\delta = \tau - |S_T|$. We have $|S_R| = |S| - |S_T| < \tau - |S_T| = \delta \leq \binom{\delta+1}{2}$ (as $\binom{\delta+1}{2} \geq \delta$ for $\delta \geq 1$), so the survival bound fails and $M(S) = 0$. $\square$

4 The crossing route, and why the prefix law needed a direct proof

The Baxterised monodromy $M(x) = \prod_k \check{R}_{i_k}(x)$, $\check{R}_i(x) = P_q^{(i)} + r(x)P_{-1}^{(i)}$, $r(x) = (q^2 - x)/(q(x - 1))$, satisfies the crossing identity $M(x)M(q^2/x) = I$ (each factor obeys $\check{R}_i(x)\check{R}_i(q^2/x) = I$ because $r(x)r(q^2/x) = 1$). One is tempted to transport the tail deficit law onto the prefix by the word-reversal that crossing induces. We record why this fails, so the route is not re-tried.

Proposition 7. *Let $M_{\text{rev}}(x)$ be the monodromy over the reversed staircase word. Then:*

- (i) *Crossing implies word-reversal invariance, $M_{\text{fwd}}(x) = M_{\text{rev}}(x)$ on V^λ for all x ; matching coefficients of $r(x)^g$ gives only the per-grade identity $\sum_{|S|=g} M(S) = \sum_{|S|=g} M_{\text{rev}}(S)$, not a per-subset statement.*
- (ii) *Independently of crossing, the seminormal generators are self-adjoint for the contravariant form, so $M(S)^* = M_{\text{rev}}(S)$; hence $M(S) = 0 \iff M_{\text{rev}}(S) = 0$ (a free per-subset symmetry).*
- (iii) *The reversed word is the ascending staircase; reversal sends the forward prefix (the short blocks $B_1, \dots, B_{n-3}$) to the end of the ascending word, whose trailing blocks are again short. The tail deficit law's engine is the long-block hook structure $B_{n-1}V^\lambda = E_{n-1}^+ = \Phi(V^{(2, 1^m)})$; the short trailing blocks carry no such structure. So neither (i) nor (ii) transports the tail law onto the prefix.*

Proof. (i) $M(q^2/x) = M(x)^{-1} = (\prod_k \check{R}_{i_k}(x))^{-1} = \prod_k \check{R}_{i_{N+1-k}}(q^2/x) = M_{\text{rev}}(q^2/x)$, so $M_{\text{fwd}} = M_{\text{rev}}$ as functions of the spectral parameter; the coefficient identity is immediate. (ii) Each $P_q^{(i)}, P_{-1}^{(i)}$ is self-adjoint for the form making the T_i self-adjoint, and adjunction reverses products. (iii) Direct inspection of the reversed word. All three are verified numerically on $(2, 2, 1, 1, 1)$ (crossing residual $< 10^{-14}$, reversal residual $< 10^{-21}$, adjoint rank-equality on all tested subsets). $\square$

Thus the prefix law is genuinely a new computation, carried out directly in §§??–§?? through the axial-gap commutation of Lemma ??.

5 Verification

All checks reuse the Hoefsmit seminormal machinery; structural facts are exact rational ($q = 5/7$), bulk sweeps are exact or high-precision numeric at generic q . Scripts in `~/projects/scratch/2026-05-30-bethe-`

- *Tail input* (??): $\text{dep}(M_T(\emptyset)V^\lambda) = \tau$ exactly for $m = 3, 4$ ($P_q^{(j)}$ annihilates the tail image for $j = 1, \dots, \tau$).
- *Lemma ??*: across the first prefix block B_{n-3} at input depth τ , the minimal- P_{-1} survivor has output depth exactly $\tau - 1$ (no survivor drops ≥ 2). Verified $m = 3, 4$.
- *Theorem ??*: the implication “ $|S_T| < \tau$ and $|S_R| < \binom{\tau - |S_T| + 1}{2} \Rightarrow M(S) = 0$ ” holds with no violation over all subsets $|S| < D$: 67 law-covered subsets for $m = 3$, 6595 for $m = 4$.
- *Corollary ??*: $M(S) = 0$ for all 56 prefix-supported subsets with $|S| < D$ at $m = 3$, by *exact* rational arithmetic; the first prefix-supported survivor occurs at $|S| = D = 3$.
- *Corollary ??*: $M(S) = 0$ for all $|S| < \tau$, exhaustively for $m = 2, 3, 4$.
- *Proposition ??*: crossing, reversal invariance and the adjoint flip confirmed numerically on $(2, 2, 1, 1, 1)$.

λ	m	τ	D	order-law slice ($S_T = \emptyset$)	ord $\geq \tau$
$(2, 2, 1, 1)$	2	1	1	first survivor $ S =1$	≥ 1
$(2, 2, 1, 1, 1)$	3	2	3	first survivor $ S =3$	≥ 2
$(2, 2, 1, 1, 1, 1)$	4	3	6	first survivor $ S =6$	≥ 3

6 Status

Proved (uniformly in m).

- Lemma ?? (a depth- d block forces d insertions) — elementary eigenprojector identity.
- Lemma ?? (a surviving block drops depth by at most one) — the only mechanism, via the axial-gap ≥ 2 commutation of generators $\geq d + 1$ past generators $\leq d - 1$.
- Theorem ?? (Prefix Deficit Law): for $|S_T| < \tau$, survival forces $|S_R| \geq \binom{\tau - |S_T| + 1}{2}$. This is the prefix counterpart requested in the Remark of the 29 May deficit paper.
- Corollary ??: the full order-law lower bound $\text{ord} \geq D$ on the slice $S_T = \emptyset$ (prefix-supported subsets).
- Corollary ??: $\text{ord}_{x=q^2} Z_\lambda \geq \tau$, all subsets.
- Proposition ??: the crossing route gives reversal invariance and a free adjoint flip, but does not transport the tail law to the prefix.

Not closed (the precise residual to the full order law $\text{ord} = D$). The Prefix Deficit Law is *exact only on the slice* $S_T = \emptyset$. The full $M(S) \equiv 0$ for all $|S| < D$ additionally requires:

- *The intermediate split.* For $1 \leq |S_T| < \tau$ the bound of Theorem ?? gives only $|S| \geq g(|S_T|)$ with $g(t) = t + \binom{\tau - t + 1}{2}$, and $g(t) < D$ for $t \geq 1$ (e.g. $g(1) = D - (\tau - 1)$). Computation shows that the subsets realising $|S| = g(t) < D$ *nonetheless vanish* (no survivor among optimal-prefix + any-tail placements at $|S| < D$ for $m = 3, 4$), so the lower bound of Theorem ??

is *not tight* for $t \geq 1$: there is extra killing, presumably from the placement of the prefix insertions relative to the sign-kill layers, that the per-block count does not yet capture.

(II) *The regime* $|S_T| \geq \tau$, outside the tail deficit law's hypothesis, which seeds the prefix analysis.

This is the same open regime flagged in the 29 May paper.

A complete proof of $\text{ord} = D$ thus needs a refinement that (a) charges the *placement* of prefix insertions, not merely their count, and (b) treats heavy tails; the present law is the exact backbone of (I) (the $S_T = \emptyset$ corner) and reduces the problem to these two residual phenomena.